

Joanna Cai

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EDUCATION

University of British Columbia, Bachelor of Applied Science in Engineering

Sept 2025 - April 2029

John Oliver Secondary School, Digital Immersion Mini School Program

Sept 2020 - June 2025

Clubs: Founder and President of Gardenbank Club, Co-President of Officials Club

TECHNICAL SKILLS

Tools and Languages: Solidworks, KiCad, Fusion 360, C, Figma, Git, Python, HTML, CSS, JavaScript, Office 365

EXPERIENCE

UBC Baja SAE Design Team

In-Person

Mechatronics Subteam Member | SolidWorks, Altium, C, Shop skills

Sept 2025 – Present

- Designed and implemented waterjet-cut linear potentiometer mounts for suspension travel measurement, enabling shock data acquisition over CAN
- Re-designed a battery box in SolidWorks to implement wire glands and secure mounting, addressing off-road vibration
- Manufactured in-house using a waterjet, drill press, welding, 3D printing, and tapping

PRIME Robotics Lab - Dr. Chao Liu

In-Person

Undergraduate Research Assistant | Solidworks, C++

Sept 2025-Present

- Designed a linear gripper in SolidWorks for reliable handling of test tubes and scoopulas in a modular chemistry automation robot.
- Researched tactile sensing, machine learning automation, and robotic manipulation while documenting proposed design decisions
- Collaborated with the principal investigator to iterate on mechanical designs, ensuring alignment with research objectives.

Nokia

Hybrid, Toronto

Software Engineering Future Tech Intern | Python, Docker, Powerpoint

July 2024 - Aug 2024

- Used Pytest and Schemathesis to write automated tests for Nokia's Notifier API.
- Communicated test results and recommendations to the security team, providing progress updates to stakeholders and pitching technical implementation with fellow interns.

PROJECTS

melokey | KiCAD, Fusion 360, Figma, Swift

June 2024 – Present WIP

- Created an electronic (MIDI) piano keyboard tailored to the needs of neurodivergent youth.
- Designed a KiCad PCB, Fusion 360 enclosure. Used Figma and Swift to design an app accompaniment.

blipyboard | KiCad, Fusion 360, Circuit Python

May 2025-June 2025

- Designed and developed a PCB, enclosure, and KMK firmware for a macro keypad with a rotary encoder and an OLED screen.

LEADERSHIP

STEM Sprouts

Vancouver, B.C.

Facilitator

June 2023 – Jan 2024

- Created a safe and inclusive environment for neurodivergent individuals aged 7-12 to access interactive STEM activities.
- Thoroughly briefed on appropriately overseeing autistic learners.

59 Vancouver Air Cadets

Vancouver, B.C.

Warrant Officer Second Class, Level 2 Warrant

Sept 2019 – June 2025

- Responsible for all level 2 cadets in their transition to level 3 as the sole aviation instructor and assistant drill instructor.
- Led squadron-wide operations for 400+ cadets promoting physical fitness, citizenship, exploration, and leadership.

AWARDS

TD Scholarship for Community Leadership

2025

- 1 of 20 young leaders selected out of 4000 national applicants.
- Full-ride scholarship valued at \$70,000 awarded to those who show a commitment to community and driving meaningful change.

Dr. Jim McEwen Innovation Award

2025

- Scholarship valued at \$5,000 presented to the student who has achieved an exceptionally innovative accomplishment, talent, skill, or ability, well beyond any expectations and requirements of courses taken by the student.